INCH POUND
MIL-PRF-1/590B
9 February 1999
SUPERSEDING
MIL-E-1/590A
30 September 1974

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBE, CATHODE RAY
TYPE 3FP7A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	V dc	v	V dc	V dc	V dc	Meg	Meg	V dc	ratio	ft
Maximum:	6.9	0	550	1,100	2,200	4,400	1.5	1.0	-125	2.3	10,000
Minimum:	5.7	-125	- - -	- - -	1,500	3,000	- - -	- - -	- - -	- - -	- - -
Test Conditions:	6.3	Adj	- - -	Focus	2,000	4,000	- - -	- - -	- - -	- - -	- - -

GENERAL:

Qualification - Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE 1. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Qualification inspection							
Neck and bulb alignment (electrostatic types)	5101	- - -		Diam	- - -	2.25	Inch
Cathode illumination	5216	- - -		- - -	- - -	- - -	- - -
Focusing voltage	5246	- - -		Eb1	400	690	V dc
Deflection factor (1D2)	5248	- - -	Eb3 = Eb2 = 2,000 V dc	DF	170	230	V dc/in.
Deflection factor (3D4)	5248	- - -	Eb3 = Eb2 = 2,000 V dc	DF	126	170	V dc/in.
Deflection factor uniformity	5248	- - -		- - -	- - -	- - -	- - -
Direct-interelectrode capacitance	1331	- - -	Control grid to all Cathode to all D1 to D2 D3 to D4 D1 to all D3 to all D1 to all except D2 D2 to all except D1 D3 to all except D4 D4 to all except D3	Cg1 Ck C1D2 C3D4 CD1 CD3 CD1 CD2 CD3 CD4	- -	10.5 10.5 4.0 4.0 15.0 12.0 13.0 13.0 12.0 12.0	pF pF pF pF pF pF pF pF pF pF
Pressure (implosion)	1141	- - -		- - -	- - -	- - -	- - -
Vibration	5111	- - -		Width	- - -	1.0	mm
Base material insulating quality	1216	- - -		- - -	- - -	- - -	- - -
Conformance inspection, part 1							
Voltage breakdown	5201	- - -		- - -	- - -	- - -	- - -
Voltage breakdown (electrostatic types)	5201	- - -		- - -	- - -	- - -	- - -
Gas “cross”	5206	1	Ib3 = 150 μA dc	- - -	- - -	- - -	- - -
Bulb, screen and faceplate quality	5106	- - -		- - -	- - -	- - -	- - -
Modulation	5223	- - -	Ib3 = 150 μA dc	ΔEc	- - -	50	V dc
Spot position (electrostatic deflection)	5231	- - -		- - -	- - -	12	mm
Spot displacement (leakage)	5231	- - -		Displ	- - -	7	mm

See footnotes at end of table.

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TABLE 1. Testing and inspection -Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Conformance inspection, part 1							
Grid cutoff voltage	5241	- - -		Eco	-30	-90	V dc
Grid No. 1 leakage current	5251	- - -		- - -	- - -	- - -	- - -
Anode No. 2 leakage current	5251	- - -		- - -	- - -	- - -	- - -
Conformance inspection, part 2							
Heater current	1301	- - -		If	540	660	mA
Electrode current (anode No. 1)	5201	- - -	Ec1 = 0	Ib1	-50	10	μA dc
Electrode current (cathode)	5201	- - -	Ib3 = 150 μA dc	Ik	- - -	1,500	μA dc
Base alignment (electrostatic types)	5101	- - -	+1D2; pin No. 5	- - -	- - -	- - -	- - -
Side terminal alignment (electrostatic types)	5101	- - -	+1D2	- - -	- - -	- - -	- - -
Angle between traces	5101	- - -		- - -	- - -	- - -	- - -
Neck and base alignment (electrostatic types)	5101	- - -		- - -	- - -	- - -	- - -
Stray light emission (conventional types)	5216	- - -	Eb2 = 2,200 V dc; Eb3 = 4,400 V dc	- - -	- - -	- - -	- - -
Screens	5221	- - -		- - -	- - -	- - -	- - -
Line width “A” (electrostatic deflection)	5226	2	Ib3 = 150 μA dc	Width	- - -	0.75	mm
Line width “B” (electrostatic deflection)	5226	3	Ib3 = 150 μA dc	Width	- - -	0.90	mm
Focusing voltage at cutoff	5246	- - -		Eb1	460	690	V dc
Deflection factor	5248	- - -	1D2	DF	212	288	V dc/in.
Deflection factor	5248	- - -	3D4	DF	153	207	V dc/in.
Heater-cathode leakage current	5251	- - -		- - -	- - -	- - -	- - -
Secureness of base, cap or insert	1101	- - -		- - -	- - -	- - -	- - -

See footnotes at end of table.

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TABLE 1. Testing and inspection -Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Conformance inspection, part 2 -Continued							
Base pin solder depth	1111	- - -		- - -	- - -	- - -	- - -
Permanence of marking	1105	- - -		- - -	- - -	- - -	- - -
Conformance inspection, part 3							
Life test	- - -	- - -	Group C; Eb3 = 4,400 V dc; Eb2 = 2,200 V dc; Ib3 = 30 μA dc; t = 500 hours	- - -	- - -	- - -	- - -
Life-test end points:	- - -	- - -					
Modulation	5223		Ib3 = 112 μA dc	ΔEc	- - -	50	V dc
Line width “A”	5226		Ib3 = 112 μA dc	Width	- - -	0.8	mm
Line width “B”	5226		Ib3 = 112 μA dc	Width	- - -	0.95	mm

NOTES:

1. This test to be performed at the conclusion of the holding period.

2. The deflecting plates shall be returned to anode number 2 through a minimum of 2.5 megohm resistors. Ib3 the beam current shall be set at 150 μA dc. The high-frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus.

   (a) The tubes shall be observed for deflection defocusing, astigmatism or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of trace (usually around edges) bow-tying (irregular widths of any single line when observed at different points) bowing of trace other than that normally caused by curvature of bulb.

   (b) This test for focus is to be made in addition to the line width measurements.

3. The same conditions shall be setup as described in note 2, except that the connection of deflecting elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of focus from maximum tube diameter in both directions without any adjustment of focus from note 2 conditions. An examination for defocusing, astigmatism or spot ellipticity shall be made as in note 2.

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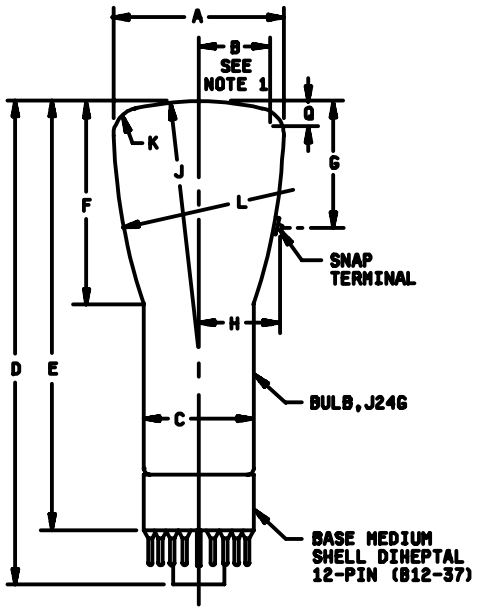

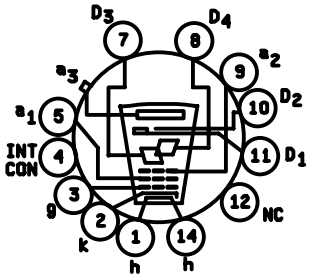

Dimensions				
Ltr	Inches		Millimeter	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	2.937 Dia.	3.063 Dia	74.60 Dia.	77.80 Dia
B	1.375		34.93	
C	1.937 Dia.	2.063 Dia	49.20 Dia.	52.40 Dia
D	9.750	10.250	247.65	260.35
G	1.500	2.000	38.10	50.80
Reference dimensions (see note 2)				
E	9.250		234.95	
F	3.875		98.43	
H	1.625		41.28	
J	8.000 Rad		203.20 Rad	
K	.250 Rad		6.35 Rad	
L	12.438 Rad		315.93 Rad	
Q	.350		8.89	

NOTES:
1. Minimum useful screen radius.
2. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube 3FP7A.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5960-3473)

Review activities:
Army - CR4, ME
Navy - AS, CG, MC, OS, SH
Air Force - 11, 17, 19, 99

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